

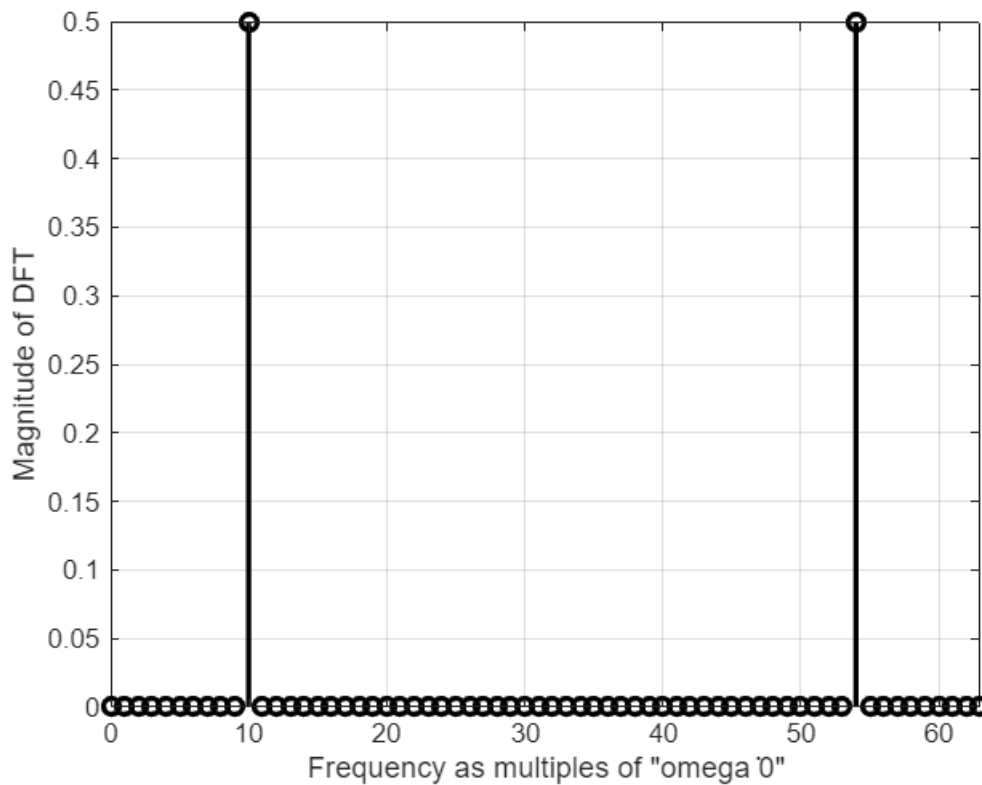
SNS Lab 10

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Example (1)

```
P = 64;  
w = 2*pi/P;  
n = 0:P-1;  
x = cos(10*w*n); %Finite-time signal  
hx = fft(x); %Compute the DFT of x  
stem(n,abs(hx)/P,'k','LineWidth',2); %Compute and plot magnitude of coefficients  
ylabel('Magnitude of DFT');  
xlabel('Frequency as multiples of "omega_0"');  
axis([0 P-1 0 inf]);  
grid;
```



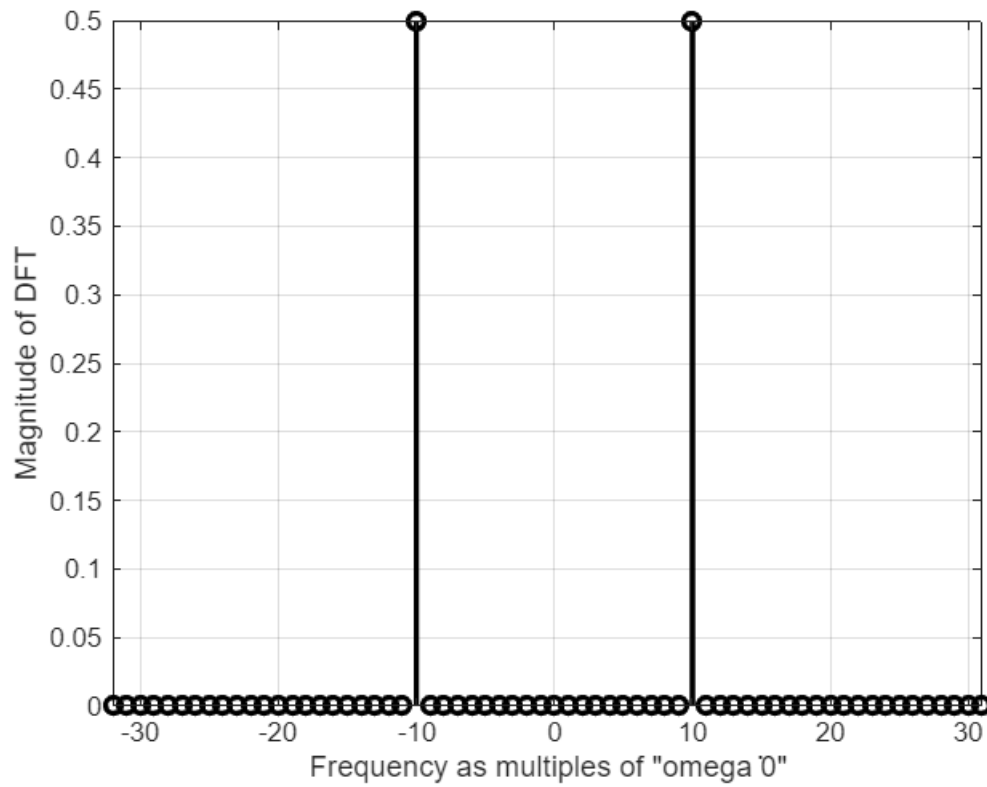
Example (2)

```
P = 64;  
w = 2*pi/P;  
n = 0:P-1;  
x = cos(10*w*n); %Finite-time signal  
hx = fft(x); %Compute the DFT of x  
shx = fftshift(hx);
```

```

stem([-P/2:P/2-1],abs(shx)/P,'k','LineWidth',2);
ylabel('Magnitude of DFT');
xlabel('Frequency as multiples of "omega_0"');
axis([-P/2 P/2-1 0 inf]);
grid;

```



Task 1(a)

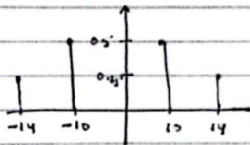
$$u[n] = \cos(10\omega_0 n) + 0.5 \cos(14\omega_0 n)$$

Using Euler's identity

$$\cos(10\omega_0 n) = \frac{e^{i(10\omega_0 n)} + e^{-i(10\omega_0 n)}}{2}$$

$$0.5 \cos(14\omega_0 n) = \frac{e^{i(14\omega_0 n)} + e^{-i(14\omega_0 n)}}{4}$$

$$u[n] = \frac{e^{i(10\omega_0 n)}}{2} + \frac{e^{-i(10\omega_0 n)}}{2} + \frac{e^{i(14\omega_0 n)}}{4} + \frac{e^{-i(14\omega_0 n)}}{4}$$



$$\omega = \frac{2\pi k}{n} \approx k = \frac{\omega n}{2\pi}$$

for 10:

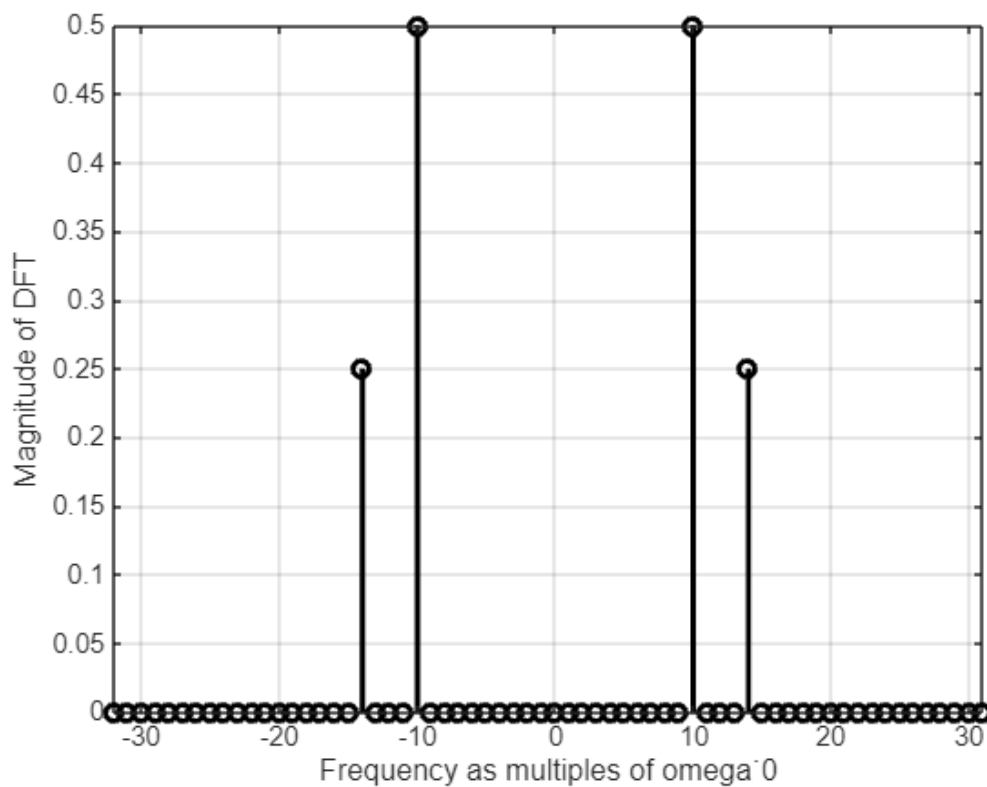
$$k = \frac{20\pi}{64} \times \frac{64}{2\pi} \Rightarrow 10$$

for 14:

$$k = \frac{28\pi}{64} \times \frac{64}{2\pi} \Rightarrow 14$$

Task 1(b)

```
P = 64;
w = 2*pi/P;
n = 0:P-1;
x = cos(10*w*n) + 0.5*cos(14*w*n); %Finite-time signal
hx = fft(x); %Compute the DFT of x
shx = fftshift(hx);
stem([-P/2:P/2-1],abs(shx)/P,"k","LineWidth",2);
ylabel("Magnitude of DFT");
xlabel("Frequency as multiples of omega_0");
axis([-P/2 P/2-1 0 inf]);
grid;
```



Task 2

```

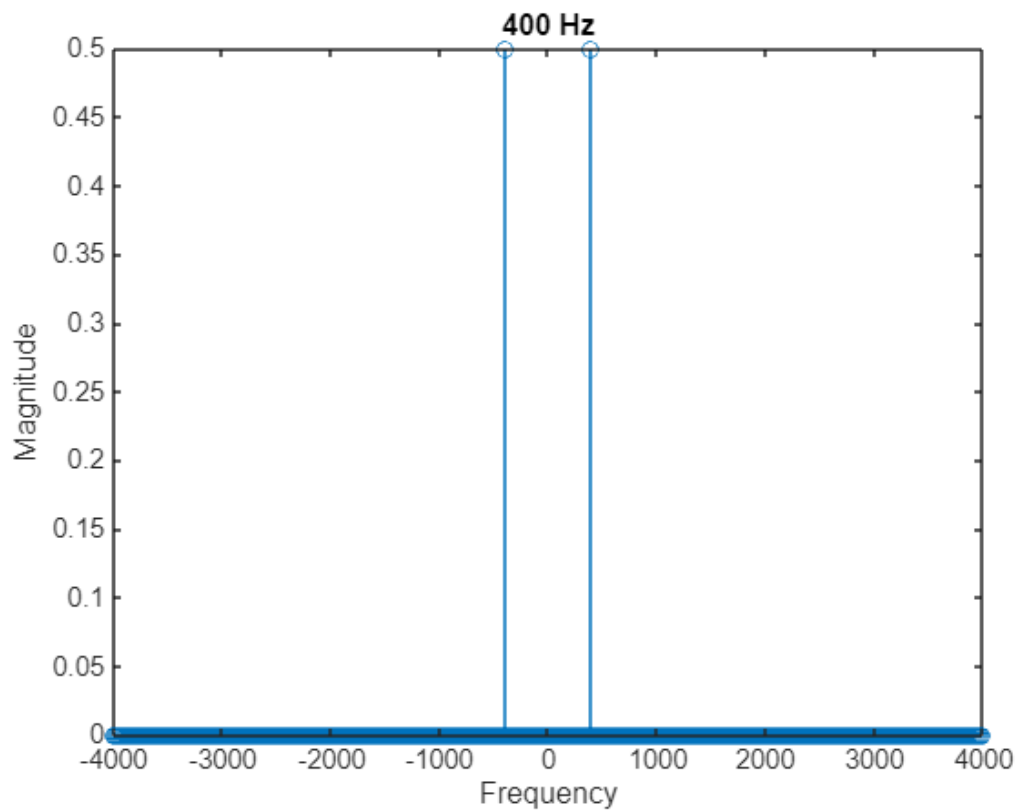
clc;
clear;
close all;

fs= 8000;
f= 400;
n= 1000;
t= (0:n-1)/fs;
x= sin(2*pi*f*t);

hx=fft(x);
shx=fftshift(hx);

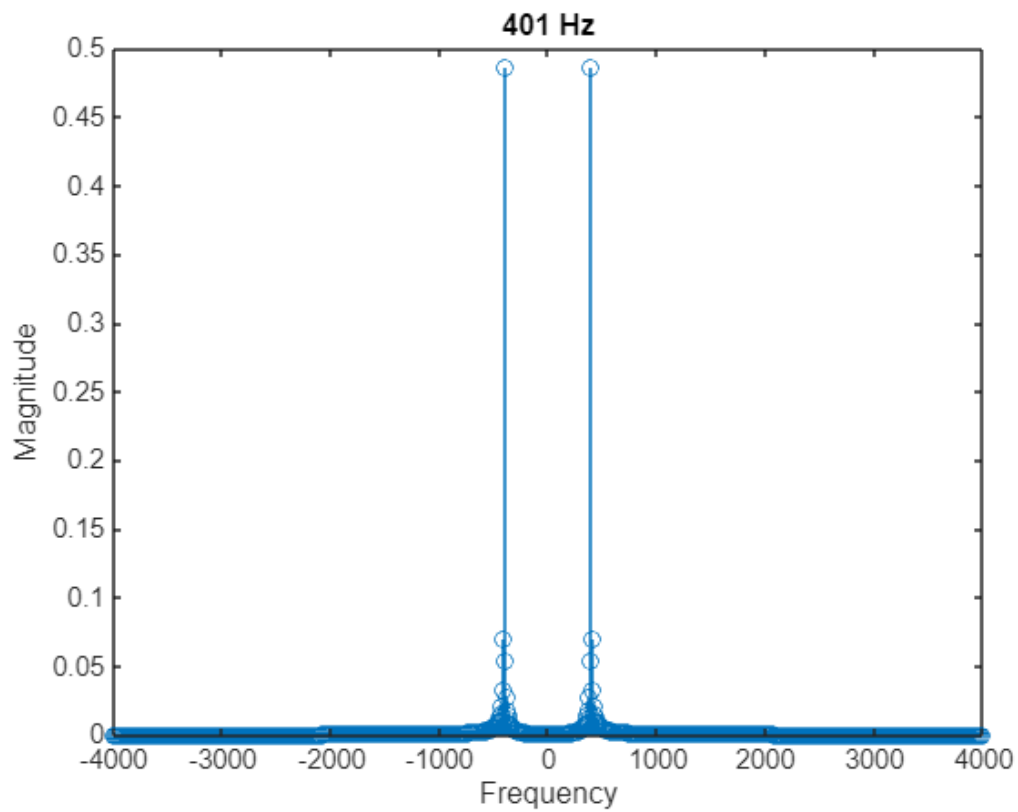
fHz= fs*(-n/2:n/2-1)/n;
figure;
stem(fHz,abs(shx)/n)
title("400 Hz");
ylabel("Magnitude");
xlabel("Frequency");

```

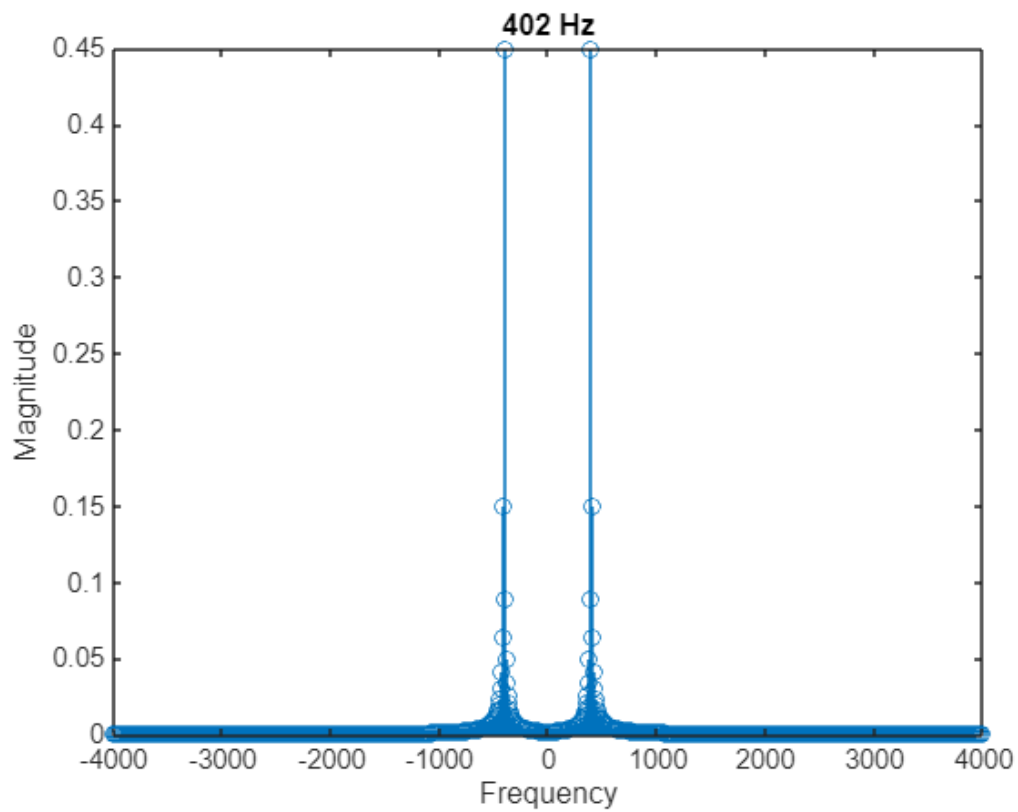


Yes the output is what we would expect it to be.

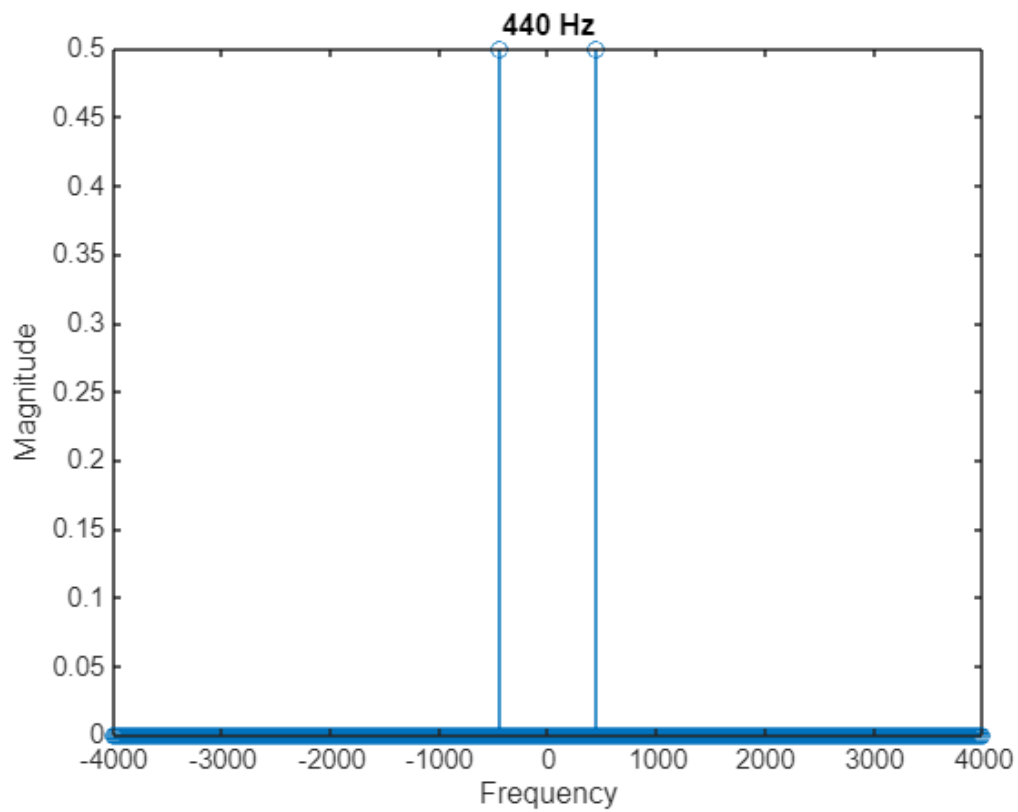
```
fs= 8000;
f= 401;
n= 1000;
t= (0:n-1)/fs;
x= sin(2*pi*f*t);
hx=fft(x);
shx=fftshift(hx);
fHz= fs*(-n/2:n/2-1)/n;
figure;
stem(fHz,abs(shx)/n)
title("401 Hz")
ylabel("Magnitude");
xlabel("Frequency");
```



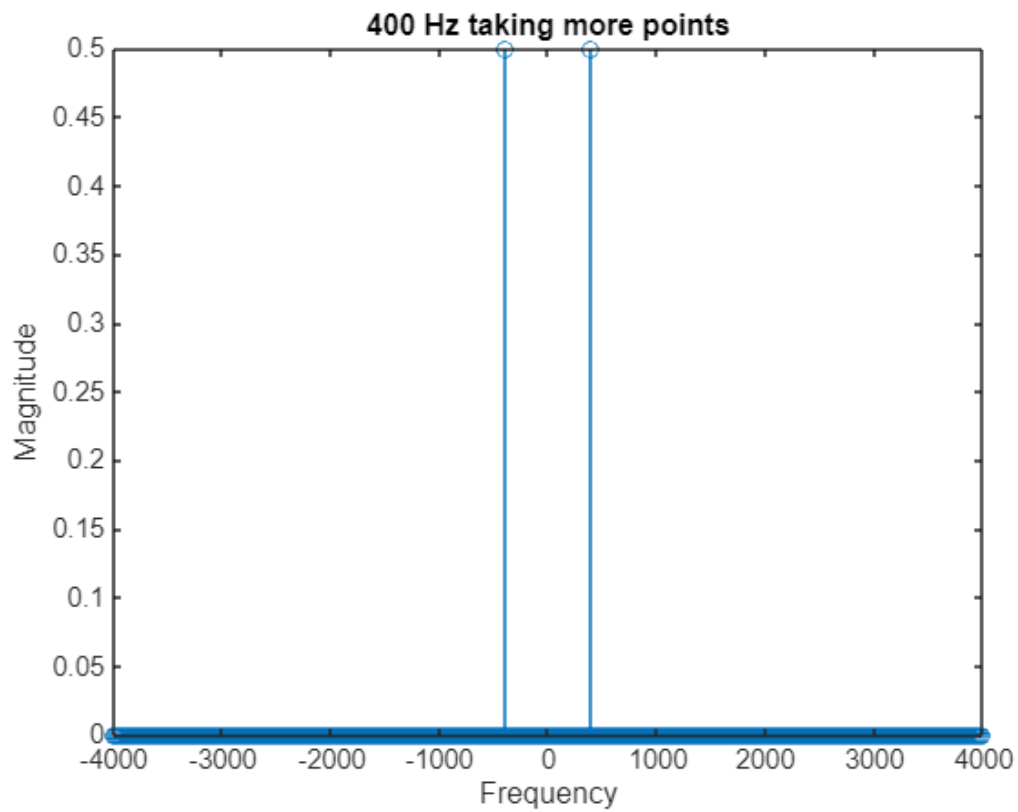
```
fs= 8000;
f= 402;
n= 1000;
t= (0:n-1)/fs;
x= sin(2*pi*f*t);
hx=fft(x);
shx=fftshift(hx);
fHz= fs*(-n/2:n/2-1)/n;
figure;
stem(fHz,abs(shx)/n)
title("402 Hz")
ylabel("Magnitude");
xlabel("Frequency");
```



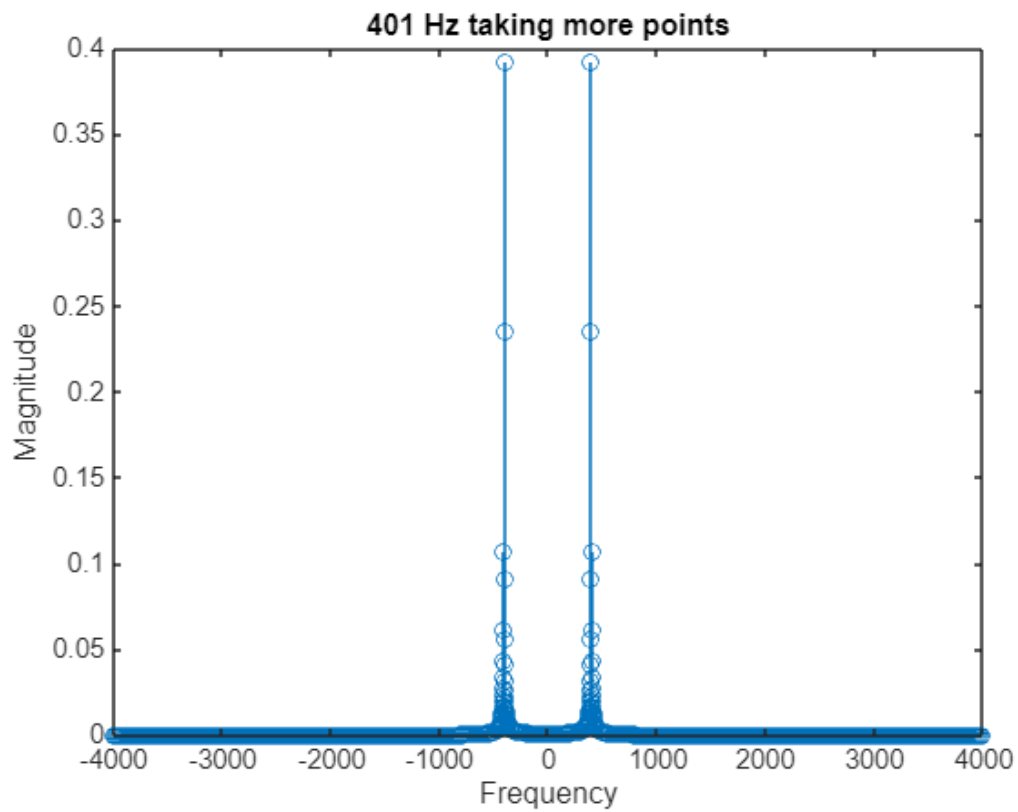
```
fs= 8000;
f= 440;
n= 1000;
t= (0:n-1)/fs;
x= sin(2*pi*f*t);
hx=fft(x);
shx=fftshift(hx);
fHz= fs*(-n/2:n/2-1)/n;
figure;
stem(fHz,abs(shx)/n)
title("440 Hz")
ylabel("Magnitude");
xlabel("Frequency");
```



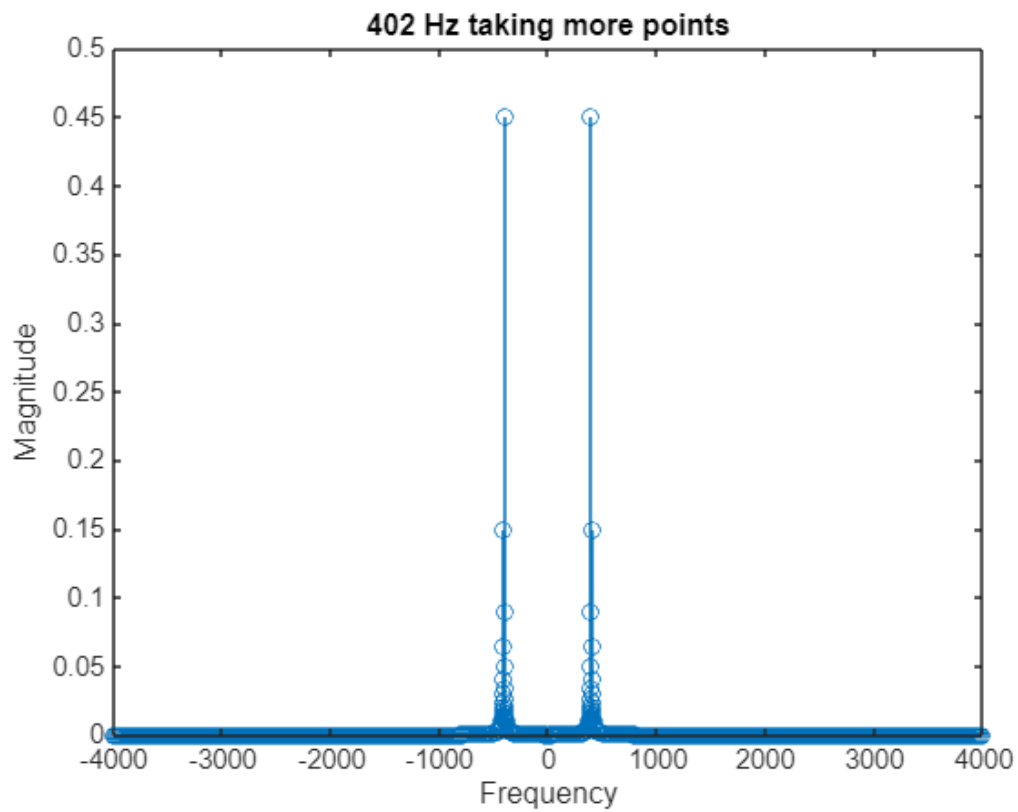
```
fs= 8000;
f= 400;
n= 5000;
t= (0:n-1)/fs;
x= sin(2*pi*f*t);
hx=fft(x,n);
shx=fftshift(hx);
fHz= fs*(-n/2:n/2-1)/n;
figure;
stem(fHz,abs(shx)/n)
title("400 Hz taking more points");
ylabel("Magnitude");
xlabel("Frequency");
```

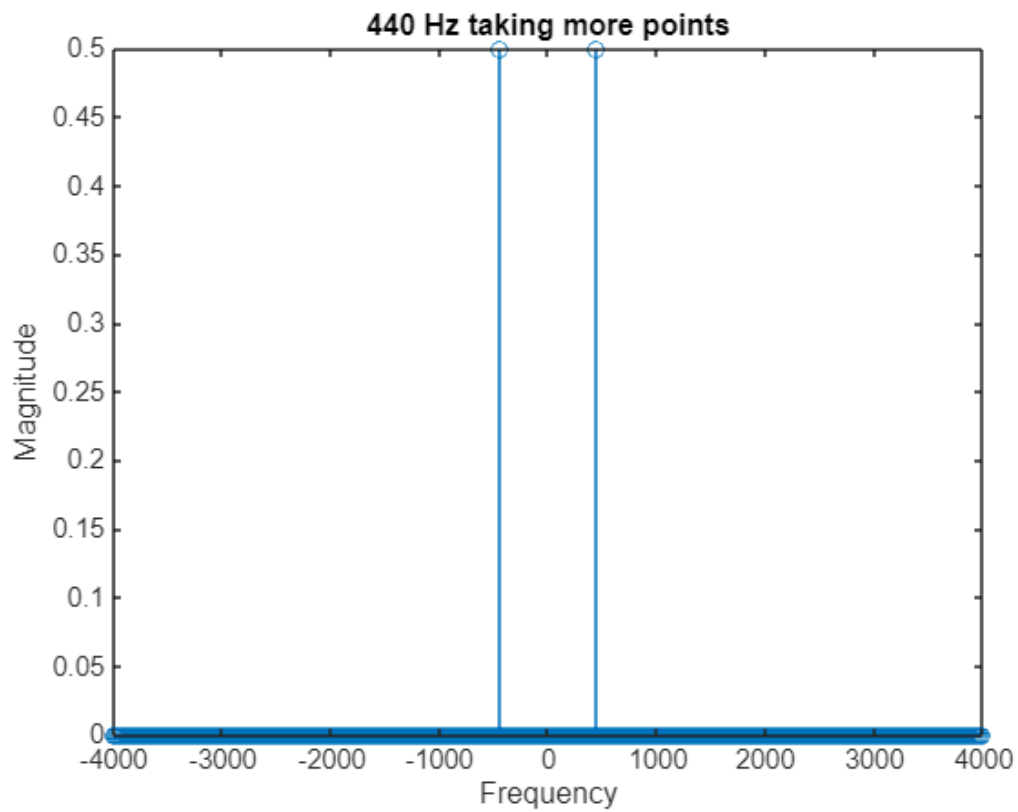
```
fs= 8000;  
f= 401;  
n= 5000;  
t= (0:n-1)/fs;  
x= sin(2*pi*f*t);  
hx=fft(x,n);  
shx=fftshift(hx);  
fHz= fs*(-n/2:n/2-1)/n;  
figure;  
stem(fHz,abs(shx)/n)  
title("401 Hz taking more points");  
ylabel("Magnitude");  
xlabel("Frequency");
```



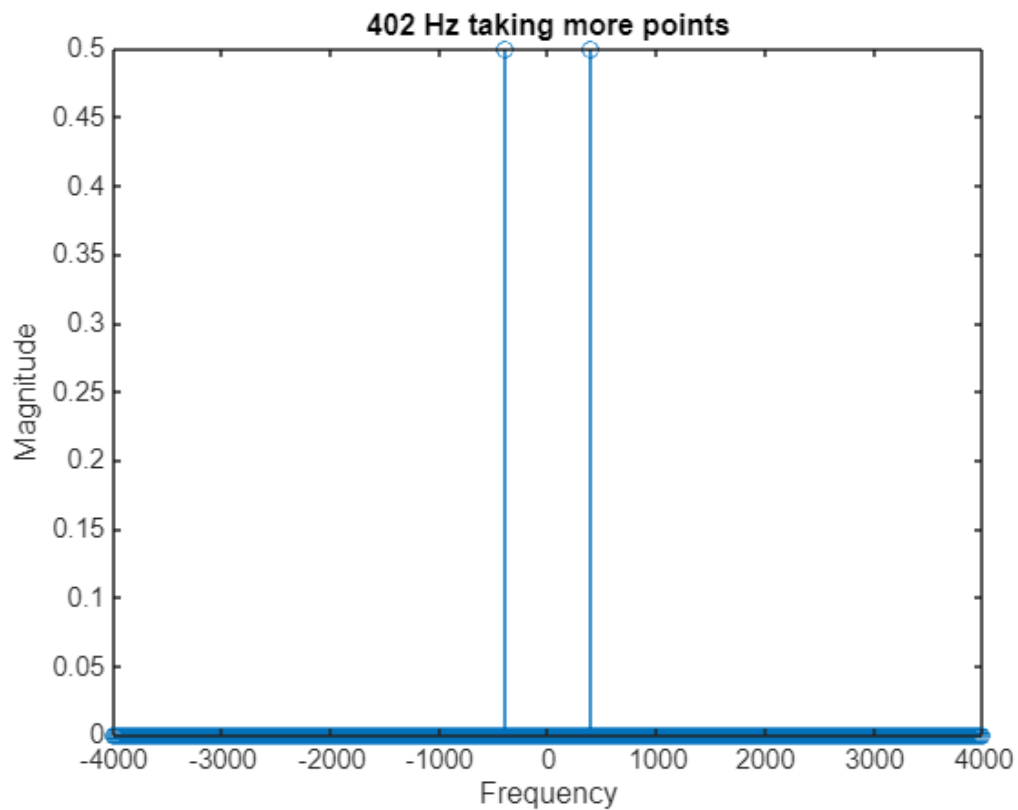
```
fs= 8000;
f= 402;
n= 5000;
t= (0:n-1)/fs;
x= sin(2*pi*f*t);
hx=fft(x,n);
shx=fftshift(hx);
fHz= fs*(-n/2:n/2-1)/n;
figure;
stem(fHz,abs(shx)/n)
title("402 Hz taking more points");
ylabel("Magnitude");
xlabel("Frequency");
```



```
fs= 8000;
f= 440;
n= 5000;
t= (0:n-1)/fs;
x= sin(2*pi*f*t);
hx=fft(x,n);
shx=fftshift(hx);
fHz= fs*(-n/2:n/2-1)/n;
figure;
stem(fHz,abs(shx)/n)
title("440 Hz taking more points");
ylabel("Magnitude");
xlabel("Frequency");
```



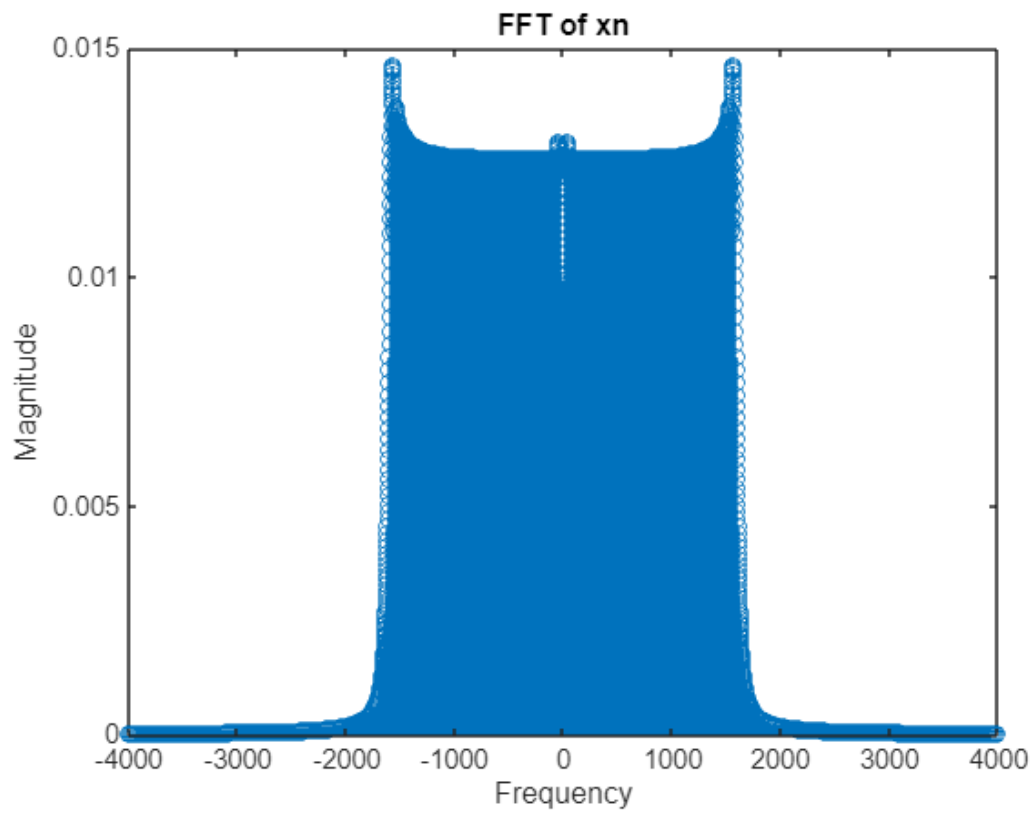
```
fs= 8000;
f= 402;
n= 4000;
t= (0:n-1)/fs;
x= sin(2*pi*f*t);
hx=fft(x,n);
shx=fftshift(hx);
fHz= fs*(-n/2:n/2-1)/n;
figure;
stem(fHz,abs(shx)/n)
title("402 Hz taking more points");
ylabel("Magnitude");
xlabel("Frequency");
```



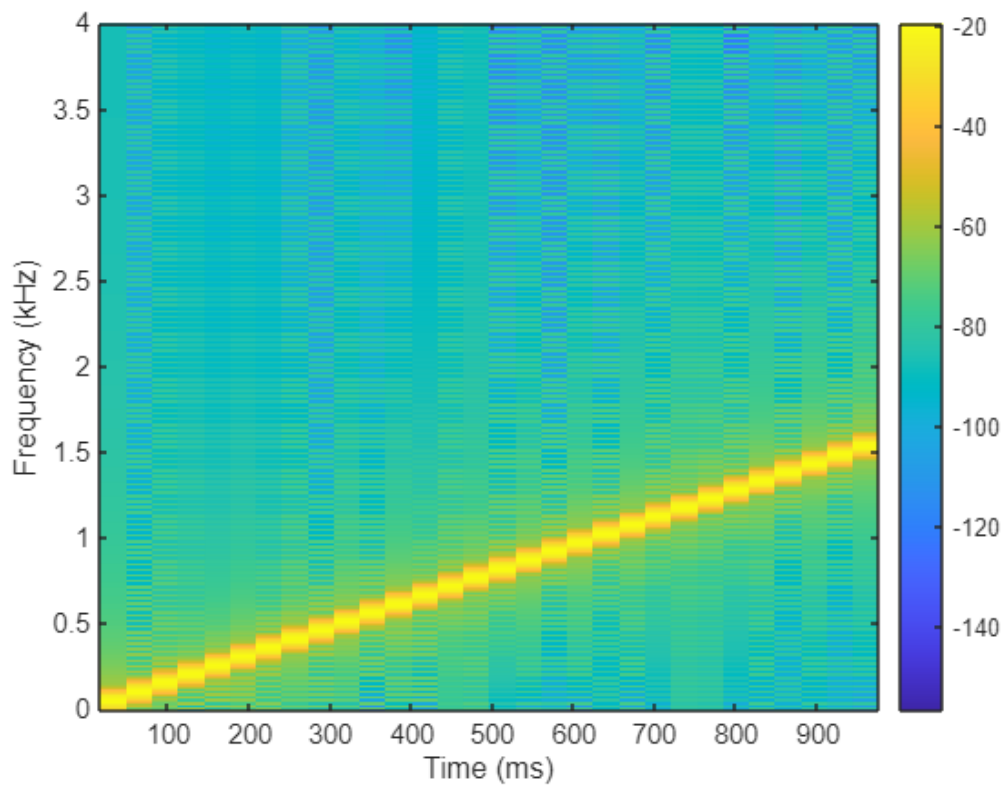
Task 3

```
clc;
clear;
close all;

fs=8000;
n=8000;
t=(0:n-1)/fs;
xn = sin(2*pi*800*t.^2);
hx=fft(xn);
shx=fftshift(hx);
fHz= fs*(-n/2:n/2-1)/n;
figure;
stem(fHz,abs(shx)/n)
title("FFT of xn");
ylabel("Magnitude");
xlabel("Frequency");
```



```
spectrogram(xn,512,[],1024,fs,"yaxis");  
colorbar
```



```
xtuu= xn(8000:-1:1);  
spectrogram(xtuu,512,[],1024,fs,"yaxis");  
colorbar
```

